Lec 17 - Math 534H

$$\int_0^\infty e^{-p^2}\,dp = \frac{\sqrt{\pi}}{2}$$

Solve:

$$\left(\int_0^\infty e^{-p^2}\,dp\right)^2 = \int_0^\infty\int_0^\infty e^{-x^2}e^{-y^2}\,dx\,dy$$

$(x,y) \to (r,\theta)$, $r = \sqrt{x^2+y^2}$

Define $S(x,t) := \frac{\partial Q(x,t)}{\partial x}$

$S_t - k S_{xx} = 0$ ← Solution

~~$u(x,y)$~~:

$$u(x,t) := \int_{-\infty}^{\infty} S(x-y,t)\,\phi(y)\,dy, \quad t>0.$$

Claim: ~~If~~ u is the unique solution to (IVP).

$$u(x,t) = \int_{-\infty}^{\infty} \frac{\partial Q(x-y,t)}{\partial x}\,\phi(y)\,dy$$

$$u(x,0) = \int_{-\infty}^{\infty} \frac{\partial Q(x-y,0)}{\partial x}\,\phi(y)\,dy = \phi(x) \ ?$$

Assume: $\phi(y) \to \infty$, $|y| \to \infty$.

$$u(x,t) = -\int_{-\infty}^{\infty} \frac{\partial}{\partial y}\left[Q(x-y,t)\right]\phi(y)\,dy$$

$\boxed{\text{Differentiable in parts}}$

$$\cancel{= \int_{-\infty}^{\infty} Q(x-y,t)\,\phi'(y) - \left[Q(x-y,t)\,\phi(y)\right]_{-\infty}^{\infty}}$$

$$= \int_{-\infty}^{\infty} Q(x-y,t)\,\phi'(y) - \left[Q(x-y,t)\,\phi(y)\right]_{-\infty}^{\infty}$$

Recall $Q(x,0) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$

$\Downarrow$

$$Q(x-y,0) = \begin{cases} 1 & x \geq y \\ 0 & x < y \end{cases}$$

$$u(x,0) = \int_{-\infty}^{\infty} Q(x-y,0)\,\phi'(y)\,dy$$

$$= \int_{-\infty}^{x} \phi'(y)\,dy = \underline{\phi(y)\Big|_{-\infty}^{x}} = \phi(x)$$

① $\lim_{y\to\infty} \phi(y) = 0$

Heat kernel

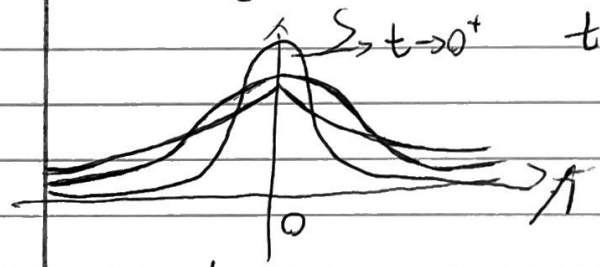
$$s(x,t) = \frac{\partial Q(x,t)}{\partial x} = \frac{1}{\sqrt{4\pi k t}}\, e^{-x^2/4kt}, \quad t > 0$$

$$u(x,t) = \frac{1}{\sqrt{4\pi k t}} \int_{-\infty}^{\infty} e^{-(x-y)^2/4kt}\,\phi(y)\,dy$$

~~Property~~ $s(x,t) = s(-x,t)$ ~~t is some fixed number~~

Property of s:

① $s(x,t) = s(-x,t)$



For larger is a squared out Gaussian, for t intermediate is a Gaussian, but for $t \to 0^+$ $s(x,t)$ spikes out $x=0$ so that in the limit you get "delta function" $\delta(x) = \begin{cases} \text{"}\infty\text{"}, & x = 0 \\ 0, & x \neq 0 \end{cases}$

generalized function / distribution

The height of these spikes (maximum of $s(x,t)$ as $t \to 0^+$ is given by $1/\sqrt{4\pi k t}$.

(2)

For large $\int_{-\infty}^{\infty} S(x,t)\,dx = 1 \quad \forall t > 0$.

~~Finally~~:

(a) The max $\underset{|x| \geq \delta}{} S(x,t) \xrightarrow[t \to 0^+]{} 0$ in fact.

$$\max_{|x| \geq \varepsilon} S(x,t) = \frac{e^{-\varepsilon^2/4kt}}{\sqrt{4\pi k t}}$$

Error function (statistics):

$$\operatorname{Erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-p^2}\,dp$$

$$\Downarrow$$

$$Q(x,t) = \frac{1}{2} + \frac{1}{2} \operatorname{Erf}\left(\frac{x}{\sqrt{4kt}}\right)$$

Class HW:

Read: Ex 2 in 2.4